MIL-F-63271A (AR)
25 July 1981
SUPERSEDING
MIL-F-63271 (AR)
28 May 1978

MILITARY SPECIFICATION

FUZE, POINT INITIATING, BASE DETONATING M509A2/METAL PARTS, WITH DETONATOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, assembly and packing for one type of artillery fuze designated as Fuze, Point Initiating, Base Detonating, M509A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation Methods of

MIL-A-13812 - Steel Plate, Wrought, Homogeneous, For Ammunition Testing (1/4 to 12 Inches, Incl.)
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-D-60031 - Detonator, Electric, M69

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331 - Fuze and Fuze Components Environmental and Performance Tests for

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8799738 - Cover, Booster
8799768 - Cup Lead
8837807 - Box for Fuze PIBD, M509A2 or Fuze PIBD, M530
9248976 - Box, Fiberboard Packing, Ammunition for Fuze, PIBD, M509A2, M530, M530A1
9323857 - Fuze PI, BD, M509A2 Mpts Assy/W/Detonator

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Fuze assembly. The fuze assembly shall comply with all requirements specified on Drawing (Dwg.) 9323857 and with all requirements specified in applicable specifications.

3.3 Arming of release mechanism assembly. The release mechanism shall arm when subjected to a maximum (max.) centrifugal acceleration (see 6.4) of 4300 g's, when tested as specified in 4.5.1.

3.4 Non-arming of release mechanism assembly. The release mechanism assembly shall not arm when subjected to a minimum (min.) centrifugal acceleration (see 6.4) of 2500 gravities (g's), when tested as specified in 4.5.2.

3.5 Arming of fuze. The metal parts assembly, with detonator, shall arm when subjected to a maximum (max.) centrifugal acceleration (see 6.4) of 4300 gravities (g's), when tested as specified in 4.5.3.

3.6 Non-arming of fuze. The metal parts assembly, with detonator, shall not arm when subjected to a minimum (min.) centrifugal acceleration (see 6.4) of 2500 gravities (g's), when tested as specified in 4.5.4.

3.7 Functioning. There shall be no evidence of premature burst and the fuze shall function the projectile high order as evidenced by complete penetration of the armor plate target by jet action, when tested as specified in 4.5.5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the current version before use.

3.8 Jolt and jumble. The metal parts assembly, with detonator, shall comply with the requirements of MIL-STD-331, Test No. 101.2, Procedure II for Jolt and Test No. 102.1 for Jumble.

3.9 Transportation vibration. The metal parts assembly, with detonator, shall comply with the requirements of MIL-STD-331, Test No. 401.

3.10 Embrittlement of springs. The springs shall not break when tested to the minimum point of permanent distortion. The test shall be performed as specified in 4.5.8.

3.11 Detonator M69. The M69 Electric Detonator used in loading these fuzes shall comply with MIL-D-60031 and the requirements of this specification.

3.11.1 Detonator deterioration. Detonators scheduled for use in loading these fuzes, shall not be used if two (2) or more years have elapsed since the last government acceptance test or if the detonators or packing/packaging, exhibits evidence of having been subjected to adverse environmental conditions, however brief, until tested for possible deterioration as specified in MIL-D-60031 and the quality verified as acceptable.

3.12 Workmanship. All parts shall be fabricated and finished in a thorough workmanlike manner. The parts shall be free of burrs, chips, cracks, sharp edges, unblended radii, surface defects, and dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.13 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.14 Data requirements. The contractor shall generate data in accordance with the data item descriptions cited in 4.3.4 and 4.4.2.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are part of this detailed specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection
b. Production Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The manufacturer of the fuzes, prior to the initiation of regular production, shall ship to a Government approved facility designated by the Contracting Officer for evaluation, an initial production sample consisting of 25 completed assemblies (with detonator) and 25 complete sets of parts (i.e., 25 each of every component and every subassembly) for evaluation in accordance with the provisions of 4.3.2. An additional eighty (80) metal parts assemblies with detonators, shields, booster covers and lead cups, which have been inspected, shall be shipped to the appropriate proving ground for acceptance functioning tests (see 3.7 and 4.5.5).

4.3.2 Inspections to be performed. The first article sample assemblies and components shall be inspected for conformance with any or all of the requirements of this specification including proving ground tests (see 4.5.5) and any or all of the requirements of the applicable drawings (see Table I).

4.3.2.1 Proving ground tests

4.3.2.1.1 Plate performance, 105mm. Twenty (20) fuzes shall be loaded and tested in Cartridge, 105MM, HEAT-T, M456 at ambient temperature as specified in 4.5.5.1.

4.3.2.1.2 Plate performance, 106mm. Twenty (20) fuzes shall be loaded and tested in Cartridge, 106MM, HEAT-T, M344A1, at ambient temperature as specified in 4.5.5.1.

4.3.2.1.3 Extreme temperature plate performance (INFORMATION ONLY). Forty (40) fuzes shall be loaded and tested for informational purposes only at extreme temperatures in accordance with 4.5.5.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the currentversion before use.

4.3.3 Rejection (See MIL-A-48078). In the ballistic tests, if a premature burst occurs or if two (2) or more projectiles fail to penetrate the plate in either the 105MM or 106MM plate performance test (not including extreme temperature information tests), the initial production sample shall be rejected.

4.3.4 Process controls and plans. The contractor shall submit, prior to the initiation of production, a written plan of process and process controls to assure that critical components contain no process defects. Included in the plan shall be specifically the process controls for the following:

a. Destructive load tests of rotor pin, leaf pins, leaf spring posts, leaf posts and leaf spring.

(1) The contractor shall define the process used to fabricate the first article quantity for the specific assemblies/parts identified. This definition shall state specifically the tools or equipment and technique used to insert the pins or posts into the base material and the instrument used to disrupt the base of the posts to assure proper integrity. The manner in which the leaf springs are coiled or processed to assure proper tension shall be specified. These process definitions may be submitted as a listing of equipment/tools complemented with a description of the primary process steps.

(2) The intended process controls to assure proper quality of the specific assemblies/parts identified shall be provided by the contractor upon submission of first article. The controls shall include, as applicable, the design force for insertion of pins, periodic inspections of hole or shaft diameter, and load deflection of the leaf spring in terms of actual value and limits anticipated. The sample size and frequency shall be specified. These shall be verified as adequate through a push out (or pull) test of a minimum of 5 pieces used to establish a positive value of integrity in terms of force. The actual values shall be recorded for information and used to evaluate the acceptability of process changes as required.

b. The contractor's written plan of process and process controls shall be submitted in accordance with Data Item Description DI-P-1604, with Attachment 1, as modified by Commander, ARRADCOM, Dover, NJ. One copy of the plan shall be prepared for each of the following recipients: Commander ARRADCOM, ATTN: DRDAR-LCU-CV and DRDAR-QAT-B, Dover, NJ 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

c. The results of the destructive load tests, as specified by the contractor's written process control plan shall be prepared for each lot in accordance with Data Item Description DI-T-1906, with copies forwarded to Commander, ARRADCOM, ATTN: DRDAR-LCU-CV and DRDAR-QAT-B, Dover, NJ 07801.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Detonators bearing the same lot interfix number from one manufacturer.

4.4.2 Examination. One hundred percent examination shall be performed for all critical defects and major defects when so specified. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plans of Standard MIL-STD-105, Level II. All nonconforming material shall be rejected. (See MIL-A-48078).

a. A summary report of all critical defects found; the number of parts or assemblies inspected for critical defects, listed by each part or assembly; and, the corrective action taken, if any, shall be prepared monthly for the duration of the contract in accordance with Data Item Description DI-R-1721, with copies forwarded to the following: Commander, ARRADCOM, ATTN: DRDAR-LCU-CV and DRDAR-QAR-E, Dover, NJ 07801.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze, Point Initiating, Base Detonating, M509A2 Detonator Parts, Assemblies and Sub-Assemblies	1 OF 6	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, lead (Dwg. 8799768)				
	Examination of Defects	25		3.2/3.12	4.4.2.1
	Cover booster (Dwg. 8799738)				
	Examination of defects	25		3.2/3.12	4.4.2.2
	Spring (Dwg. 8799824)				
	Examination of defects	25		3.2/3.12	4.4.2.3
	Embrittlement Test	25		3/2/3.12	4.58
	Rear bearing plate (Dwg. 8799820)				
	Examination of defects	25		3.2/3.12	4.4.2.4
	Anti-reset spring (Dwg. 8799819)				
	Examination of defects	25		3.2/3.12	4.4.2.5
	Front bearing plate (Dwg. 8799785)	25		3.2/3.12	4.4.2.6
	Examination of defects	25		3.2/3.12	4.4.2.6

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 6		DRAWING NUMBER
	Fuze, Point Initiating, Base Detonating, M509A2 Detonator Parts, Assemblies and Sub-Assemblies (Continuation Sheet)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Leaf Number 3 (Dwg. 8799818)				
	Examination of defects	25		3.2/3.12	4.4.2.7
	Release mechanism assembly (Dwg. 8799777)				
	Arming test, Non-arming test				
	Examination of defects				4.4.2.8
	Torque of bearing plate screws	25		3.2./3.12	4.5.1 & 4.5.2
	Spring, rotor (Dwg. 8799773)				
	Examination of defects	25		3.2/3.12	4.4.2.9
	Embrittlement test	25		3.2/3.12	4.5.8
	Shield (Dwg. 9323850)				
	Examination of defects	25		3.2/3.12	4.4.2.10
	Shield assembly (Dwg. 9323858)				
	Examination of defects	25		3.2/3.12	4.4.2.11
	Nut speed, Push-on (Dwg. 9323832)				
	Examination of defects	25		3.2/3.12	4.4.2.12

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	CONTINUATION SHEET	3 OF 6	See below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Insulator contact (Dwg. 9323830)				
	Examination of defects	25		3.2/3.12	4.4.2.13
	Spring contact (Dwg. 9323827)				
	Examination of defects	25		3.2/3.12	4.4.2.14
	Embrittlement Test	25		3.2/3.12	4.5.8
	Contact (Dwg. 9323829)				
	Examination of defects	25		3.2/3.12	4.4.2.15
	Housing contact after splitting (Dwg. 9323828)				
	Examination of defects	25		3.2/3.12	4.4.2.16
	Contact assembly (Dwg. 9323875)				
	Examination of defects	25		3.2/3.12	4.4.2.17/9323875
	Detonator and contact assembly (Dwg. 9323833)				
	Examination of defects	25		3.2/3.12	4.4.2.18/9323833

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 6		DRAWING NUMBER
	CONTINUATION SHEET				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rotor detonator and contact assembly (Dwg. 9323882)				
	Examination of defects Detonator and contact ass'y load test Contact button load when compressed	25		3.2/3.12	4.4.2.19/4.5.15/4.5.16
	Rotor (Dwg. 9323822)				
	Examination of defects	25		3.2/3.12	4.4.2.20
	Rotor and pin assembly (Dwg. 9323851)				
	Examination of defects	25		3.2/3.12	4.4.2.21/4.5.14
	Bushing, contact Dwg. 9323846)				
	Examination of defects	25		3.2/3.12	4.4.2.22
	Sleeve, insulator (Dwg. 9323979)				
	Examination of defects	25		3.2/3.12	4.4.2.23
	Assembly, housing contact (Dwg. 9323847)				
	Examination of defects	25		3.2/3.12	4.4.2.24

NOTES:

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]
Check the source to verify that this is the current version before use.

TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
	CONTINUATION SHEET				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Housing, rotor (Dwg. 9323852)				
	Examination of defects	25		3.2/3.12	4.4.2.25
	Washer, conducting prior to plating (Dwg. 9323854)				
	Examination of defects	25		3.2/3.12	4.4.2.26
	Washer, conducting after plating (Dwg. 9323854)				
	Examination of defects	25		3.2/3.12	4.4.2.27
	Housing assembly (Dwg. 9323911)				
	Examination of defects Resistance between conducting washer & housing less than 62,000 ohms or greater than 200,000 ohms	25		3.2/3.12	4.4.2.28/4.5.13
	Housing and contact bushing assembly (Dwg. 9323853)				
	Examination of defects Load test failure of contact bushing	25		3.2/3.12	4.4.2.29/4.5.12

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 6 OF 6		DRAWING NUMBER
	CONTINUATION SHEET				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal parts assembly w/detonator (Dwg. 9323857)				
	Examination of defects	25		3.2/3.12	4.4.2.30
	Arming	25	Note 1	3.5	4.5.3
	Non-Arming	25	Note 1	3.6	4.5.4
	Jolt and Jumble	12	Note 1	3.8	4.5.6
	Transportation Vibration	13	Note 1	3.9	4.5.7
	Plate Performance	80	Note 2	3.7	4.5.5

NOTES: 1. Use a sample of completed assemblies with detonators which have been examined for defects in accordance with 4.4.2.30.
2. To be performed at the designated Government proving ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Cup, lead		8799768
			NEXT HIGHER ASSEMBLY
			8799735

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Split or cut completely thru wall		100%	3.2	Visual
MAJOR					
101	Distance from bottom to flange		0.40%	3.2	Gage
102	Thickness thru bottom		0.40%	3.2	Gage
103	Thickness of flange		0.40%	3.2	Gage
104	Evidence of poor workmanship		0.40%	3.12	Visual
MINOR					
201	Diameter of flange		0.65%	3.2	Gage
202	Diameter under flange		0.65%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cover, Booster	1 OF 1	8799738
			NEXT HIGHER ASSEMBLY: 8799735

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.2	Gage
MINOR					
201	Length		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.12	Visual
204	Thickness thru bottom		0.65%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Spring	1	1	8799824
				NEXT HIGHER ASSEMBLY: 8799819

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Thickness, min.		0.40%	3.2	Gage
102	Location of rivet holes		0.40%	3.2	Gage
103	Length from bottom surface to flat		0.40%	3.2	Gage
104	Crack		0.40%	3.2	Visual
105	Embrittlement test		Note 1	3.10	Test/4.5.8
MINOR					
201	Diameter of rivet holes, max.		0.65%	3.2	Gage
202	Width, min.		0.65%	3.2	Gage
203	End radii missing or incorrect		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: Note 1: See 4.4.3.6 for sampling plan.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Rear Bearing Plate		8799820
			NEXT HIGHER ASSEMBLY 8799777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Location of leaf pin holes with relation to rotor pivot hole		100%	3.2	Gage
MAJOR					
101	Location of anti-reset spring assembly holes		0.40%	3.2	Gage
102	Location and length of rotor pin slot		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8799819
4.4.2.5	Anti-reset Spring Assembly				NEXT HIGHER ASSEMBLY 8799777
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length from rotor shaft hole to spring		0.40%	3.2	Gage
102	Length from leaf pin holes to end of spring		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Front Bearing Plate	1 OF 1	8799785
			NEXT HIGHER ASSEMBLY 8799777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Location of leaf pin holes with relation to rotor pivot hole.		100%	3.2	Gage
MAJOR	None defined				
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

18

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Leaf Number 3		8799818
			NEXT HIGHER ASSEMBLY
			8799777

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Profile of restraining point in relation to pivot hole		100%	3.2	Gage
MAJOR	None defined				
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Release Mechanism Assembly	1 OF 1	8799777
			NEXT HIGHER ASSEMBLY: 9323857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Spring ends not in leaf grooves		100%	3.2	Visual
2	Leaf groove missing		100%	3.2	Visual
3	Leaf number 1 inverted		100%	3.4	Visual
4	Non-arming Test failure		100%	3.4	Test/4.5.2
MAJOR					
101	Anti-reset spring assembly missing		0.40%	3.2	Visual
102	Torque of bearing plate screws		0.40%	3.2	Test/4.5.10
103	Arming test failure		100%	3.3	Test/4.5.1
MINOR					
201	Gap between bearing plate and stud spacer		0.65%	3.2	Visual
202	Spring pin or stud inverted		0.65%	3.2	Visual
203	Plating damaged exposing base metal		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Spring, Rotor				8799773
					NEXT HIGHER ASSEMBLY
					9323857
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Distance from coil center to hook, min.		0.40%	3.2	Gage
102	Load at assembled position		0.40%	3.2	Visual
103	Crack or bend		0.40%	3.2	Visual
104	Embrittlement Test		Note 1	3.10	Test/4.5.8
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: Note 1: See 4.4.3.6 for sampling plan.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.10	Shield		9323850
			NEXT HIGHER ASSEMBLY
			9323858

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Depth of release mechanism cavity		0.40%	3.2	Gage
102	Depth of booster pellet cavity		0.40%	3.2	Gage
103	Thickness through bottom of release mechanism cavity		0.40%	3.2	Gage
104	Diameter of booster cavity		0.40%	3.2	Gage
105	Diameter of booster lead cup hole		0.40%	3.2	Gage
106	Diameter of booster lead cup hole counterbore		0.40%	3.2	Gage
107	Perpendicularity of front face of release mechanism cavity with outside diameter		0.40%	3.2	Gage
108	Perpendicularity of booster cavity front face with outside diameter		0.40%	3.2	Gage
109	Diameter over knurl, min.		0.40%	3.2	Gage
110	Depth from bottom of lead cup counterbore to bottom of release mechanism cavity, min.		0.40%	3.2	Gage
111	Outside diameter		0.40%	3.2	Gage
112	Location of holes in body from front face of release mechanism cavity (4 places)		0.40%	3.2	Gage
113	Angular location of holes in body (4 places)		0.40%	3.2	Gage
114	Location of fuze position pin hole		0.40%	3.2	Gage
115	Diameter of fuze positioning pin hole		0.40%	3.2	Gage
116	Inside diameter, max.		0.40%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	CONTINUATION SHEET	2 OF 2	9323850
			NEXT HIGHER ASSEMBLY: 9323858

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Chamfer on booster end missing		0.65%	3.2	Visual
202	Plating damaged exposing base metal		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Shield Assembly	1 OF 1	9323858
			NEXT HIGHER ASSEMBLY: 9323859

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Spring pin missing		0.40%	3.2	Visual
102	Protrusion of pin above outside surface		0.40%	3.2	Gage
MINOR					
201	Spring pin above flush with outside surface		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9323832
4.4.2.12	Nut, Speed, Push-on				NEXT HIGHER ASSEMBLY 9323875
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Diameter, max.		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Insulator, Contact	1 OF 1	9323830
			NEXT HIGHER ASSEMBLY
			9323875

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Outside diameter, min.		0.40%	3.2	Gage
102	Inside diameter, max.		0.40%	3.2	Gage
103	Bottom wall thickness, max.		0.40%	3.2	Gage
104	Diameter of hole, max.		0.40%	3.2	Gage
105	Total length		0.40%	3.2	Gage
106	Inside radius, min.		0.40%	3.2	Gage
107	Outside radius, max.		0.40%	3.2	Gage
108	Inside surface finish improper		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Spring Contact				9323827
					NEXT HIGHER ASSEMBLY 9323875
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Outside diameter, max. (IN. .164 hole)		0.40%	3.2	Gage
102	Inside diameter, min. (over .115 rod)		0.40%	3.2	Gage
103	Gold flash incomplete		0.40%	3.2	Visual
104	Solid height, max.		0.40%	3.2	Gage
105	Spring pressure when compressed		0.40%	3.2	Gage
106	Ends not closed		0.40%	3.2	Visual
107	Embrittlement Test		Note 1	3.10	Test/4.5.8
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES: Note 1: See 4.4.3.6 for sampling plan.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Contact	1 OF 1	9323829
			NEXT HIGHER ASSEMBLY 9323875

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length from shoulder to end of contact, min.		0.40%	3.2	Gage
102	Length of large diameter, min.		0.40%	3.2	Gage
103	Radius over max.		0.40%	3.2	Gage
104	Inside diameter, max.		0.40%	3.2	Gage
105	Large diameter of drill point, max.		0.40%	3.2	Gage
106	Large outside diameter		0.40%	3.2	Gage
107	Small outside diameter		0.40%	3.2	Gage
108	Length of inside diameter		0.40%	3.2	Gage
109	Finish improper		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Housing, Contact After Splitting and Forming End	1 OF 1	9323828
			NEXT HIGHER ASSEMBLY: 9323875

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Small inside diameter, max.		0.40%	3.2	Gage
102	Large inside diameter, max.		0.40%	3.2	Gage
103	Large outside diameter		0.40%	3.2	Gage
104	Length of large inside diameter		0.40%	3.2	Gage
105	Radius improper		0.40%	3.2	Gage
106	Protective finish thickness, min.		0.40%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-28T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9323875
4.4.2.17	Contact Assembly				NEXT HIGHER ASSEMBLY 9323833
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length from base of insulator to tip of contact		0.40%	3.2	Gage
102	Outside diameter of housing contact, max.		0.40%	3.2	Gage
103	Contact fails to operate freely		0.40%	3.2	Manual
104	Electrical contact failure between contact and housing		100%	3.2	Test (9323875)
105	Crimp not full 360 degrees		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Detonator and Contact Assembly		9323833
			NEXT HIGHER ASSEMBLY 9323882

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Electrical contact failure between contact button and detonator cup		0.40%	3.2	Test (Dwg 9323833)
102	Total length, min.		0.40%	3.2	Gage
103	Contact fails to operate freely		0.40%	3.2	Manual
104	Crimp not bottomed against detonator		0.40%	3.2	Visual
105	Red lacquer missing from output end of detonator		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Rotor, Detonator and Contact Assembly	1 OF 1	9323882
			NEXT HIGHER ASSEMBLY: 9323857

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Height of contact button, min.		0.40%	3.2	Gage
102	Detonator and contact assembly load test		0.40%	3.2	Test 4.5.15
103	Contact button load when compressed to specified height		0.40%	3.2	Test (Dwg. 9323882)
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 20261026T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.20	Rotor				9323822
					NEXT HIGHER ASSEMBLY
					9323851
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Perpendicularity of rotor pin hole with face of large boss		0.40%	3.2	Gage
102	Diameter of rotor pin hole counterbore, max.		0.40%	3.2	Gage
103	Angular location of rotor pin hole		0.40%	3.2	Gage
104	Diameter of small boss, rotor pivot pin		0.40%	3.2	Gage
105	Diameter of rotor pivot pin		0.40%	3.2	Gage
106	Width of spring slot in rotor pivot pin		0.40%	3.2	Gage
107	Concentricity of rotor pivot pin with outside diameter of rotor		0.40%	3.2	Gage
108	Outside diameter of rotor		0.40%	3.2	Gage
109	Thickness of rotor body		0.40%	3.2	Gage
110	Total height		0.40%	3.2	Gage
111	Depth of large portion of detonator and contact assembly hole		0.40%	3.2	Gage
112	Depth to bottom of small portion of detonator		0.40%	3.2	Gage
113	Diameter of detonator and contact assembly through hole		0.40%	3.2	Gage
114	Location of detonator and contact assembly hole		0.40%	3.2	Gage
115	Diameter of small portion of detonator and contact assembly hole		0.40%	3.2	Gage

NOTES:

Source: https://assist.dla.mil - Downloaded: 2022-Dec-27 04:04 MST
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.20	CONTINUATION SHEET				9323822
					NEXT HIGHER ASSEMBLY 9323851
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Diameter of large portion of detonator and contact assembly hole		0.40%	3.2	Gage
117	Concentricity between small diameter and large diameter of detonator and contact assembly hole		0.40%	3.2	Gage
118	Position of rotor spring slot		0.40%	3.2	Gage
119	Height of large boss		0.40%	3.2	Gage
120	Finish improper on rotor diameter or rotor bottom		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Rotor and Pin Assembly		9323851
			NEXT HIGHER ASSEMBLY 9323882

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Load test failure of pin		100%	3.2	Test - 4.5.14
2	Length rotor pin protrudes, min.		100%	3.2	Gage
MAJOR					
101	Length rotor pin potrudes, max.		100%	3.2	Gage
102	Perpendicularity of pin diameter with face of rotor		100%	3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-28T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Bushing, Contact	1 OF 1	9323846
			NEXT HIGHER ASSEMBLY 9323847

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length		0.40%	3.2	Gage
102	Thickness of head, min.		0.40%	3.2	Gage
103	Diameter over knurl, min.		0.40%	3.2	Gage
104	Pitch diameter of thread, max.		0.40%	3.2	Gage
105	Minor diameter of thread, max.		0.40%	3.2	Gage
106	Length of thread, min.		0.40%	3.2	Gage
107	Diameter of head, min.		0.40%	3.2	Gage
108	Thickness of coating		0.40%	3.2	Test
MINOR					
201	Pitch diameter of thread, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.23	Sleeve, Insulator		9323979
			NEXT HIGHER ASSEMBLY
			9323847

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Large inside diameter, max.		0.40%	3.2	Gage
102	Small inside diameter, max.		0.40%	3.2	Gage
103	Total length, max.		0.40%	3.2	Gage
104	Length of large outside diameter, max.		0.40%	3.2	Gage
105	Large outside diameter, max.		0.40%	3.2	Gage
106	Inside radius, max.		0.40%	3.2	Gage
107	Outside chamfer		0.40%	3.2	Gage
108	Crack		0.40%	3.2	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2025-Feb-26 04:41 MST
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.24	Assembly, Housing Contact		9323847
			NEXT HIGHER ASSEMBLY 9323853

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length contact bushing extends beyond insulator sleeve		0.40%	3.2	Gage
102	Length from top of contact bushing to end of large portion of insulator sleeve		0.40%	3.2	Gage
103	Insulator sleeve cracked or damaged		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.25	Housing, Rotor		9323852
			NEXT HIGHER ASSEMBLY
			9323853

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Diameter of rotor cavity		0.40%	3.2	Gage
102	Perpendicularity of top face with small outside diameter		0.40%	3.2	Gage
103	Location of contact bushing hole		0.40%	3.2	Gage
104	Depth of rotor cavity		0.40%	3.2	Gage
105	Location of rotor contact slot		0.40%	3.2	Gage
106	Width of rotor contact slot		0.40%	3.2	Gage
107	Depth to bottom of rotor contact slot from top surface		0.40%	3.2	Gage
108	Location of rotor cavity		0.40%	3.2	Gage
109	Location of screw holes		0.40%	3.2	Gage
110	Total length of housing		0.40%	3.2	Gage
111	Diameter of contact bushing hole counterbore, max.		0.40%	3.2	Gage
112	Diameter of contact bushing hole, max.		0.40%	3.2	Gage
113	Concentricity of contact bushing hole with diameter of contact bushing hole counterbore		0.40%	3.2	Gage
114	Depth to bottom bleeder resistor cavity from rear surface		0.40%	3.2	Gage
115	Length from top surface to flange		0.40%	3.2	Gage
116	Location of spring stud and spring pin holes		0.40%	3.2	Gage
117	Diameter of shield staking holes		0.40%	3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 256 Feb 71 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.25	CONTINUATION SHEET				9323852
					NEXT HIGHER ASSEMBLY
					9323853
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
117	Diameter of shield staking holes		0.40%	3.2	Gage
118	Location of shield staking holes		0.40%	3.2	Gage
119	Radius at bottom of contact bushing hole counterbore, max.		0.40%	3.2	Gage
120	Finish of rotor cavity improper		0.40%	3.2	Visual
MINOR					
201	Diameter of housing under flange, min.		0.65%	3.2	Gage
202	Diameter of flange		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

40

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.26	Washer, Conducting Prior to Plating		9323854
			NEXT HIGHER ASSEMBLY
			9323911

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Thickness		0.40%	3.2	Gage
MINOR	None defined				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74, Which is Obsolete

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Washer, Conducting, After Plating				9323854
					NEXT HIGHER ASSEMBLY
					9323911
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Plating thickness		0.40%	3.2	Gage
102	Inside diameter, max.		0.40%	3.2	Gage
MINOR	None defined				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.28	Housing Assembly				9323911
					NEXT HIGHER ASSEMBLY 9323857
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length from top surface to contact bushing assy		100%	3.2	Gage
102	Load test of contact bushing assembly		0.40%	3.2	Test/4.5.11
103	Resistance between conducting washer and housing less than 62,000 ohms or greater than 200,000 ohms		100%	3.2	Test 4.5.13
104	Staking missing or incomplete		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

43

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil - Downloaded: 2020-10-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.29	Housing and contact bushing assembly				9323853
					NEXT HIGHER ASSEMBLY
					9323911
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Length from forward end to contact bushing assembly under load		100%	3.2	Test 4.5.11
MAJOR					
101	Torque test of housing contact		0.40%	3.2	Test 4.5.12
102	Insulation resistance of rotor housing and contact bushing		100%	3.2	Test 4.5.17
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Metal Parts Assembly, with Detonator	1 OF 1	9323857
			NEXT HIGHER ASSEMBLY: 8799735

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Rotor in armed position		100%	3.2	Visual
2	Conducting washer missing		100%	3.2	Visual
3	Insulator resistor assembly missing		100%	3.2	Visual
4	Clearance between the outside edge of the rear bearing plate rotor pin slot and the restraining tip of leaf number 3, max.		100%	3.2	Gage
5	Rotor spring not hooked under lug		100%	3.2	Visual
MAJOR					
101	Release mechanism assembly screw missing or loose		100%	3.2	Visual-Manual
MINOR					
201	Diameter of flange, max.		0.65%	3.2	Gage
202	Total height, max.		0.65%	3.2	Gage
203	Component plating damaged exposing base metal		0.65%	3.2	Visual
204	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2020-10-27T04:01Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.31	Interior packing carton, prior to sealing	1 OF 1	9248976
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Packing component missing		0.40%	3.2	Visual
102	Incorrect quantity of fuzes		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	Seal interior packing carton	1 OF 1	9248796
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Carton damaged to the extent contents may be exposed		0.40%	3.2	Visual
102	Contents loose		0.40%	3.2	Visual
103	Marking missing, misleading or unidentifiable		0.40%	3.2	Visual
104	Sealing strip missing or insecure		0.40%	3.2	Visual-Manual
MINOR					
201	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

47

Source: https://assist.dla.mil - Downloaded: 2026-02-26T04:26Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63271A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.33	Box, packing				8837807
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Box damaged exposing contents		0.40%	3.2	Visual
102	Strapping missing, broken or loose		0.40%	3.2	Visual-Manual
103	Contents loose		0.40%	3.2	Manual
MINOR					
201	Strapping improperly assembled		0.65%	3.2	Visual
202	Marking missing misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.12	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.4.3 Testing

4.4.3.1 Arming of fuze - major defect (see 3.5). This test shall be performed 100 percent. The test shall be performed as specified in 4.5.3, using test equipment specified in 4.4.4 any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

4.4.3.2 Non-arming of fuze - critical defect (see 3.6). Immediately after having successfully passed the arming test specified in 4.4.3.1 above, the fuze shall be reset to the unarmed position and subjected to the non-arming test specified in 4.5.4, using test equipment specified in 4.4.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot and the finding immediately reported.

4.4.3.3 Functioning

4.4.3.3.1 Plate performance (see 3.7). Premature burst - Critical defect, plate penetration failure - Major defect - These tests shall be performed at a proving ground in accordance with 4.5.5 (see 6.7).

4.4.3.3.1.1 Plate performance in cartridge, 105mm, HEAT-T, M456

4.4.3.3.1.2 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirement, forty (40) assemblies shall be randomly selected from each lot for this test. The lot shall be rejected if a premature burst occurs or if three (3) or more projectiles fail to function and penetrate the plate.

4.4.3.3.1.3 After three (3) consecutive lots have complied with 4.4.3.3.1.2 above, twenty (20) assemblies shall be selected at random from each lot for this test. If four (4) or more projectiles fail to function and penetrate the plate the lot shall be rejected. If two (2) or three (3) projectiles fail to function and penetrate the plate, a second sample of forty (40) fuze assemblies shall be selected and tested. If the total number of failures in the first and second sample combined is 4 or more the lot shall be rejected.

4.4.3.3.2 Plate performance in cartridge, 106mm, HEAT, M344A1

4.4.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirement,

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the current version before use.

forty (40) assemblies shall be selected at random from each lot for this test. The lot shall be rejected if a premature burst occurs or if three (3) or more projectiles fail to function and penetrate the plate.

4.4.3.3.2.2 After three (3) consecutive lots have complied with 4.4.3.3.2.1 above, twenty (20) assemblies shall be selected at random from each lot for this test. If four (4) or more projectiles fail to function and penetrate the plate the lot shall be rejected. If two (2) or three (3) projectiles fail to function and penetrate the plate, a second sample of forty (40) fuze assemblies shall be selected and tested. If the total number of failures in the first and second sample combined is 4 or more the lot shall be rejected. If a premature burst occurs at any time, the lot shall be rejected.

4.4.3.3.3 Extreme temperature plate performance (FOR INFORMATION ONLY). Forty (40) fuze assemblies shall be selected at random from each of the first three (3) lots for this informational test. The test shall be performed as specified in accordance with 4.5.5.2.

4.4.3.4 Jolt and jumble (see 3.8). Critical defects. A sample of thirty-six (36) assemblies shall be selected from each lot for these tests. The assemblies shall be subjected to the jolt test specified in MIL-STD-331, except that after test, the fuze assemblies shall not be disassembled but shall be immediately subjected to the jumble test specified in MIL-STD-331. The assemblies shall then be disassembled to determine compliance with the applicable requirement. If any assembly fails to comply with the requirement, the lot shall be rejected. Assemblies subjected to these tests shall not be returned to the lot.

4.4.3.5 Transportation vibration (see 3.9). Defect classificaton is as shown in the following:

Critical Defect	Major Defect
1. Breakage of, damage to or loosening of any part likely to result in a safety hazard during Transportation Vibration (TV) Test	1. Failure to arm after Transportation Vibration Test
2. Detonator functioning during Transportation Vibration Test	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

3. Assembly Arms during Transportation Vibration Test

4. Failure of Non-arming Test after Transportation Vibration Test

4.4.3.5.1 A sample of thirty-six (36) assemblies shall be selected from each lot for this test. The assemblies shall be subjected to the test specified in MIL-STD-331 and 4.5.7. The fuze assemblies shall then be subjected to the Non-arming test specified in 4.5.4. If any fuze fails to comply with the non-arming requirement (3.6) the lot shall be rejected. If all 36 fuze assemblies pass the Non-arming test, the same fuzes shall then be subjected to the Arming test specified in 4.5.3. If one or more assemblies fail to comply with the arming requirement (see 3.5), the lot shall be rejected. If any assembly or component fails to comply with the requirements of MIL-STD-331, the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot.

4.4.3.6 Embrittlement of springs (see 3.10 and see dwgs 8799781, (spring leaf), 8799824 (spring), 8799773, (spring rotor), and 9323827 (spring contact). Major Defects. A sample of thirty (30) springs of each type shall be selected from each heat treat batch of springs and tested as specified in 4.5.8. If one or more springs fail to meet the applicable requirement the lot of springs represented by the sample shall be rejected.

4.4.3.7 Load test of leaf No. 2 assembly (dwg 8799814 and Leaf No. 3 assembly (dwg. 8799817) Major Defects. A sample of 300 of each type of leaf assembly shall be selected from each lot for these tests. If one or more assemblies fail to comply with the specified dwg. requirement the lot represented by that leaf shall be rejected. The test shall be performed as specified in 4.5.9, using test equipment specified in 4.4.4.

4.4.3.8 Resistance of insulator resistor assembly. (See dwg. 8799752 Notes 4 and 6). A sample of 300 insulator resistor assemblies shall be selected from each lot for these tests. The tests shall be performed as specified on the applicable drawing using test equipment specified in 4.4.4. If one or more resistors fail to comply with the specified drawing requirements the lot shall be rejected (see 6.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Arming test of release mechanism assembly. The release mechanism assembly, with rotor and pin assembly and rotor spring assembled, shall be mounted in a centrifuge with the longitudinal axis parallel to the rotating arm with the bearing plate screws pointing away from the center of rotation. The centrifuge shall then be rotated at the required arming speed and then stopped and the assembly visually inspected for arming by means of a 3 to 5 power magnification to determine arming by the following means:

a. The assembly shall be considered armed when the Leaf Number 3 assembly has rotated sufficiently to release the rotor pin and permit the rotor to turn until the rotor pin is in contact with the end of the slot in the rear bearing plate.

b. The spring of the anti-reset spring assembly locks against Leaf Number 3. All assemblies which arm shall be reset in the non-armed condition and then subjected to the non-arming test.

4.5.2 Non-arming test of release mechanism assembly. The release mechanism assembly, with rotor and pin assembly and rotor spring assembled, shall be mounted in a centrifuge with the Longitudinal axis parallel to the rotating arm with the bearing plate screws pointing away from the center of rotation. The centrifuge shall then be rotated at the required non-arming speed and then stopped and the assembly visually inspected for non-arming by means of a 3 to 5 power magnification to determine non-arming by the following criteria:

a. The assembly shall be considered unarmed when the rotor pin is still restrained by the Leaf restraining point of Leaf Number 3 (Attempt to release Leaf Number 3 by pushing on the restraining point with a wooden probe) (see para. 4.4.2.8 Critical #4).

b. The pins in Leafs Number 2 and 3 are on the forward flat portions of Leafs Number 1 and 2 respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the currentversion before use.

c. The Leaf Number 3 engagement under the anti-reset spring shall be approximately 3/64 inch. If engagement is approximately 1/64 inch or less, the release mechanism shall be classed defective.

4.5.3 Arming. The fuze metal parts assembly with detonator and contact assembly in unstaked shields shall be mounted in a centrifuge with the longitudinal axis parallel to the rotating arm and the flanged end of the fuze pointing away from the center of rotation. The centrifuge shall then be rotated at the required arming speed and then stopped and the assembly inspected for arming visually and checked electrically as follows:

a. The Leaf Number 3 assembly has rotated sufficiently to release the rotor pin and permit the rotor to turn until the rotor pin is in contact with the end of the slot in the rear bearing plate.

b. The spring of the anti-reset spring assembly locks against Leaf Number 3.

c. The resistance between the conducting washer and the rotor housing is not less than 1000 ohms nor more than 8600 ohms. The resistance measuring instrument shall have a current output of not more than 20 microamperes and an error of not more than 2 percent for the full scale. This test shall be performed behind a suitable barricade. Any assembly which fails to arm shall be classed defective. All assemblies which successfully arm shall be subjected to the Non-arming test.

4.5.4 Non-arming. This test shall be performed as specified in 4.5.3 except that the centrifuge shall be operated at the non-arming speed, then stopped and the assembly removed and inspected for evidence that the fuze assembly is secured in the non-arming position. Such evidence shall be achieved by 100 percent examination by means of a 3 to 5 power magnification to determine non-arming by the following criteria.

a. Visual examination for rotor in the unarmed position.

b. Rotor pin held firmly in position by the restraining point of leaf Number 3 (attempt to release rotor by pushing the point with a wooden probe).

c. The pins in leafs Number 2 and 3 are on the forward flat portions of leafs 1 and 2 respectively.

d. The leaf Number 3 engagement under the anti-reset spring shall be approximately 3/64 inch. If engagement is approximately 1/64 inch or less, replace the release mechanism, assemble and retest the fuze assembly for arming and non-arming.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.5.5 Functioning. These tests shall be performed at a Government Proving Ground.

4.5.5.1 Plate performance. The fuze assemblies with leads and booster pellet, shall be assembled into the M456 and M344A1 cartridges as specified in 4.3.2.1 as applicable. The cartridges shall be fired at ambient temperatures at 5-inch homogeneous armor plate placed in a vertical position approximately 400 feet from the muzzle of the gun. In the case of the M344A1 only, if 5-inch armor plate is not available, 4-inch, 3-inch, or 2-inch plate may be substituted. In the case of the M456 cartridge only, if 5-inch plate is not available, 6-inch plate may be substituted. Observation shall be for fuze functioning and penetration by jet action. All armor plate shall meet the requirements of MIL-A-13812.

4.5.5.2 Extreme temperature plate performance. One half of the fuzes selected for this informational test shall be loaded and assembled into Cartridge, 105MM, HEAT-T, M456 and the remainder shall be loaded and assembled into Cartridge, 106MM, HEAT, M344A1. One half of the cartridges of each type shall then be conditioned for a minimum period of 24 hours at -35°F degrees F. (plus or minus 5 degrees F.) and the remainder at 125 degrees F. (plus or minus 5 degrees F.). The cartridges shall be fired at 5-inch homogeneous armor plate placed in a vertical position approximately 400 feet from the muzzle of the gun. In the case of the M344A1 only, if 5-inch plate is not available, 4-inch, 3-inch or 2-inch plate may be substituted. Observation shall be for fuze functioning and penetration by jet action. All armor plate shall meet the requirements of MIL-A-13812.

4.5.6 Jolt and jumble. The metal parts assemblies with shield assembled and staked shall be tested in accordance with the procedures specified in MIL-STD-331 except that after completion of the jolt test the assemblies shall not be disassembled but shall be subjected to the jumble test and then disassembled to determine compliance with applicable requirements. The assemblies subjected to these tests shall not be returned to the lot.

4.5.7 Transportation vibration. The metal parts assemblies, with shield assembled and staked shall be tested in accordance with the procedures specified in MIL-STD-331 for compliance with the requirements. The assemblies subjected to these tests shall not be returned to the lot.

4.5.8 Embrittlement of springs. Springs subjected to these tests shall not be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.5.8.1 Leaf spring. Using any suitable fixture for holding the sample springs by the two coils, both legs of the spring shall be bent outward until they are aligned with that portion of the spring which joins the coils. The spring shall not break.

4.5.8.2 Anti-reset spring, spring rotor, spring contact. Using any suitable fixture for holding the sample springs, they shall be bent to the point of minimum permanent distortion. The springs shall not break.

4.5.9 Load test of leaf No. 2 and leaf No. 3. The appropriate leaf assembly shall be placed in a suitable holding fixture and the specified load shall be applied slowly in the required direction. A maximum travel rate of load application of .125 inches per minute is recommended. The assembly shall be considered a failure if the leaf dog pulls out, loosens, or if the dimensional requirements of dwg. 8799814 (Leaf No. 2 Ass'y) or dwg 8799817 (Leaf No. 3 Ass'y) are not met after test. Assemblies which meet the requirements of this test may be returned to the lot provided the components have not been damaged.

4.5.10 Torque test of bearing plate assembly screws. The minimum specified torque shall be applied in a tightening direction while holding the bearing plate assembly in a suitable fixture. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.5.11 Contact bushing load test. The housing and contact bushing assembly shall be placed in a holding fixture and the specified load shall be slowly applied in the required direction, to the contact bushing assembly. A maximum travel rate of load application of .125 inches per minute is recommended. Assemblies subjected to these tests may be returned to the lot provided the components have not been damaged.

4.5.12 Torque test of housing contact assembly. The housing and contact bushing assembly shall be placed in a holding fixture and the specified torque applied to the housing contact assembly in a clockwise direction. Any housing contact assembly which rotates shall be classed defective. Assemblies subjected to these tests may be returned to the lot provided the components have not been damaged.

4.5.13 Resistance test of housing assembly. The resistance shall be measured using the housing as one contact and the conducting washer as the other contact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

4.5.14 Rotor and pin assembly load. The rotor and pin assembly shall be placed in a suitable holding fixture and the specified load shall be slowly applied in the required direction. A maximum travel rate of load application of .125 inches per minute is recommended. The assembly shall be considered a test failure if the pin pulls out, loosens, or if the dimensional requirements of dwg. 9323851 are not met after test. Assemblies which meet the requirements of this test may be returned to the lot provided the components have not been damaged.

4.5.15 Load test of detonator and contact assembly. The rotor detonator and contact assembly shall be placed in a suitable holding fixture and the specified load applied to the detonator in the required direction. Any assembly which shows evidence of movement of the detonator and contact assembly in the rotor shall cause the assembly to be classed defective.

4.5.16 Contact button load. The rotor detonator and contact assembly shall be placed in a suitable holding fixture and the tip of the contact shall then be compressed to the specified dimension (dwg. 9323882) and the load required to maintain the dimension shall be recorded.

4.5.17 Insulation resistance of housing and contact bushing. The insulation resistance between the rotor housing and the contact bushing (less insulator resistor assembly and conducting washer) shall be measured with resistance measuring instrument that will supply an alternating current potential of 1000 volts root mean square at 60 hertz.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level B. The fuze metal parts assembly with detonator and shield assembly shall be preserved and packaged in accordance with dwg. 9248976.

5.2 Packing

5.2.1 Fuze metal parts assembly with detonator and shield assembly.

5.2.1.1 Level B. The fuze metal parts assembly with detonator and shield assembly shall be packed in accordance with dwg. 8837807.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:26Z
Check the source to verify that this is the currentversion before use.

5.2.2 Booster cover and lead cups

5.2.2.1 Level B. The booster covers and lead cups shall be packed as specified in Standard MIL-STD-1169.

5.3 Marking. Unit and exterior containers for Fuze Metal Parts Assembly shall be marked in accordance with dwgs. 8837807 and 9248976. Marking of containers for booster covers and lead sups shall be in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Intended use. The fuze covered by this specification is intended for use in Cartridge, 105MM, HEAT-T, M456 and Cartridge, 106MM, HEAT, M344A1.

6.2 Ordering data. (See 6.1 of MIL-A-48078 (PA).

6.2.1 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3.4.b	Manufacturing Methods Report	DI-P-1604 (Modified) w/Attachment 1
4.3.4.c	Test and Demonstration Report	DI-T-1906
4.4.2.a	Quality Inspection Defect Report	DI-R-1721

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

6.3 Submission of design approval. See 6.2 of MIL-A-48078. Submit equipment designs, as required to Commander, ARRADCOM, HQ., Dover, NJ 07801, ATTN: DRDAR-QAR-E.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

6.4 Centrifugal acceleration. The centrifugal acceleration expressed in gravities (g's) is calculated as follows:

$$a = .1022r(n^2)$$

where:

a = acceleration expressed in g's
r = distance in inches from the axis of the centrifuge to the center of the release mechanism (see Note).
n = revolutions per second of the centrifuge

NOTE - The distance between the center of the rotor shaft and the center of the release mechanism is 0.218 inch.

6.5 Cost of check tests. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test on Government furnished material. The test shall be conducted at Government expense without cost to the contract or assembling the fuze, and shall not consititute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the detonators were under the jurisdiction of either contractor.

6.6 Submission of test data and ammunition data cards. In addition to the normal distribution of records, copies of all ballistic test data and ammunition data cards will be forearded to the following: Commander, ARRADCOM, HQ, Dover, NJ 07801, ATTN: DRDAR-QAR-E.

6.7 Penetration. A "complete penetration" is defined as a hole produced by jet action extending completely through the plate. The lodging of a copper slug in the completed hole is not considered cause for rejection. Penetration produced by jet action is characterized by a relatively smooth tapered hole having a larger entrance than exit. A "partial penetration" occurs when a hole produced by jet action does not extend completely through the plate. The term "dud" shall be applied to rounds which merely break-up on the armor target, those which deflagrate (function low order), or those which penetrate by kinetic energy. Rounds which deflagrate are characterized by a dish shaped cavity on the front surface of the armor which may or may not be accompanied by spalling of the rear surfce. Another possible indication of a dud is a large hole, approximately the diameter of the projectile, which has been punched through the plate by kinetic energy of the projectile with no evidence of jet action. Test reports shall distinguish between duds which merely break-up with no evidence of explosion or burning and duds which deflagrate or penetrate by kinetic energy.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:26Z
Check the source to verify that this is the currentversion before use.

6.8 Resistance of insulator resistor assembly test. Insulator resistor assemblies subjected to the resistance test of dwg. 8799752 shall not be returned to the lot.

Custodian:
Army-AR

Preparing Activity
Army-AR

Project Number: 1390-A218

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-63271A FUZE, POINT INITIATING, BASE DETONATING M509A2/METAL PARTS, WITH DETONATOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:26Z
Check the source to verify that this is the current version before use.